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# REM Assignment -3

## Q.1

→ Solar Collector
A solar Collector Absorbs ~~co~~ solar radiation and convert it into thermal energy.

Types of Solar collectors
Flat plate collectors (FPC)
Used for low temperature applications components, absorber plate, tubes, glass cover.
Applications: Domestic water heater.

Evacuated tube Collectors (ETC)
Higher efficiency than FPC., Reduces heat loss, Suitable for colder regions.

Concentrating Collectors.
focus Sunlight Using mirrors & lenses
Types :- Parabotic through
- Solar Dish
- Solar tower
Used for power generation
The working principle of solar photovoltaic Systems with ~~heat dia~~

→ Solar PV system works on the photovoltaic effect where Sunlight falling on a p-n junction Semi-conducter generates DC electricity.

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main components: Solar panel, charge controller, Battery, Inverter, load grid

Conclusion: Solar PV systems provide clean, decentralised and scalable electricity solutions.

Q5)
→ solar thermal energy utilizes solar radiation to produce heat energy various application

Solar water heating
Domestic and industrial hot water
Hospitals, hotels, hostels.

Solar Cooking
Box type and parabolic cookers.
Reduces LPG Consumption

Solar Drying
Drying agricultural products.
Improves food preservation.

Industrial process heat
textile dairy food processing.
Low and medium temperature Processes.